How Temperatures Below 5°C Affect Household Gas Consumption

Marharyta Nikulina,2074109

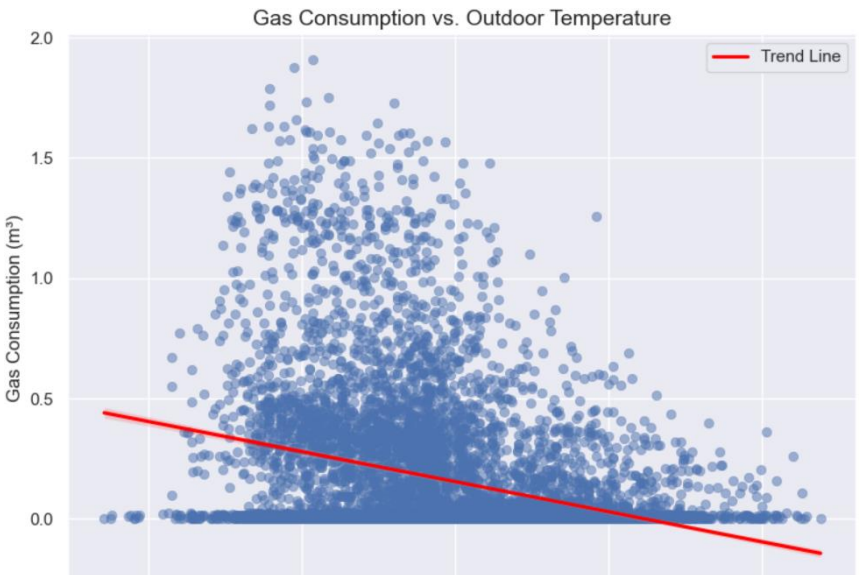
Introduction

In this study, we examine the effects of outdoor temperature (for temperatures below 5 °C) on household natural gas consumption related to heating. Some early observations of the data indicated a substantial increase in the use of gas when the temperature fell below 5°C, validating the decision to utilize 5°C as a threshold. It all comes together with a refined hypothesis — that households consume more gas on days colder than 5 degrees Celsius — directly speaking to the problem of energy demand rising in cold conditions. Being aware of this interaction is what informs energy efficiency approaches and helps households control how much they consume.

Methods

To analyze the impact of outdoor temperature on household gas consumption, a scatter plot and a bar plot were used. The scatter plot illustrates the relationship between temperature and gas usage, showing that consumption rises as temperatures drop below 5°C. The bar plot highlights the average gas consumption for colder and warmer days, making the trend more apparent. Additionally, a two-sided hypothesis test confirmed that gas consumption is significantly higher on colder days, with a very low p-value .

Results



In this plot gas consumption rises with outdoor temperature lowering. Trend line does prove so too.

This plot indeed shows that with higher outdoor temperature average gas consumption is lower.

Two-sample t-test provided following information:

The null hypothesis stated that that the consumption of gas is the same for outdoor temperature being below 5°C and above 5°C , while alternative hypothesized higher consumption on colder days. P-values turned out to be 3.21e-157 and 3.90e-157 for household 1 and household 2 respectfully, strongly confirming the rejection of null hypothesis, providing significant evidence that gas consumption does rise on colder days for both of the households.

Discussion

T-tests and various plots were applied to analyze the relationship between gas usage and outdoor temperature, revealing clear trends. Nonetheless, incorporating additional factors such as wind speed, cloud cover, and other weather conditions could provide a more thorough understanding of gas consumption patterns. Including these variables would help capture more subtle dependencies and potentially reveal more complex relationships that influence gas usage.

Conclusion

The findings underscore the effect of temperature on heating demand by demonstrating that households use more gas on colder days. This knowledge can help shape policies that increase energy efficiency and lower usage in the winter. These findings highlight the need for improved insulation and heating systems to reduce energy use during extreme temperatures as energy prices climb.